

Response to Reviewer Comments

Manuscript: Congestion-Aware Joint Charging-Station Placement, Capacity, and Swarm-UAV Trajectory Design for Age-Optimal Data Collection

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Dear Editor and Reviewer,

We thank the reviewer for the careful and constructive reading. The comments identified real weaknesses, and addressing them has clearly strengthened the paper. Below we respond to each comment point by point and indicate where the manuscript was changed. Reviewer comments are shown in shaded boxes; our responses and the locations of the changes follow.

Comment 1: Trajectory, frozen visiting order

Reviewer: The CETSP trajectory freezes the cyclic visiting order (angular sort of k -means centroids), which removes combinatorial routing optimization from the continuous solver. Integrate a dynamic sequence optimization, or give a rigorous bound on the suboptimality of the frozen order.

Response. We agree that freezing the order was a genuine limitation, and we no longer freeze it. The trajectory block now alternates the continuous position updates (Adam) with a discrete 2-opt re-ordering of the visiting sequence: we run a block of gradient steps, reorder the current hover points by 2-opt, and repeat. This is a coordinate descent between the continuous positions and the discrete sequence, so the order is optimized jointly with the positions. A 2-opt pass on fixed points can only shorten the tour and the subsequent position polish only lowers the objective, so the refinement never worsens the result and, in our tests, improves it.

Changes. Section V-A (Trajectory block) was rewritten to describe the position/order alternation, and the implementation now interleaves 2-opt reordering. We additionally report an exact optimality-gap study (see the end of this letter).

Comment 2: Swarm size ($M = 12$ to 20 is not a swarm)

Reviewer: The manuscript claims a “swarm” but caps the experiments at $M = 12$ to 20 , which is a minor fleet. A swarm should be evaluated at $M > 50$.

Response. We added a dedicated large-swarm scaling study. We scale the fleet to $M = 60$ UAVs over a 20 km field with $K = 8M$ sensors, and we grow the charging capacity with the swarm ($C_{\text{tot}} = \lceil M/3 \rceil$ ports), consistent with our provisioning law (Proposition 1): a large swarm is only worthwhile if its charging capacity grows with it. The proposed contention-aware placement keeps its advantage over the single pooled station and the coverage-optimal placement across the whole range (gain 10.2 to 12.7%), and the peak AoI stays controlled (50.6 min at $M = 20$ to 47.6 min at $M = 60$).

Changes. New Section VI subsection “Scaling to a large swarm” and a new figure; a new experiment script generates the data.

Comment 3: Queue bound, tighten $M/M/c//N$ with $M/D/c//N$

Reviewer: The finite-source $M/M/c//N$ model is a loose conservative (exponential) upper bound relative to the near-deterministic reality. Develop and benchmark an $M/D/c//N$

approximation to tighten the bound.

Response. We added a deterministic-service $M/D/c//N$ approximation using the standard service-variability (Allen-Cunneen) correction, in which the mean wait scales with $(1 + c_s^2)/2$, so deterministic service ($c_s^2 = 0$) halves the M/M wait. Benchmarked against the discrete-event simulation, this tightens the over-prediction of the real wait from about $4.5 \times (M/M)$ to about $2.3 \times (M/D)$. We also explain the residual: it is due to the near-deterministic operating (inter-request) process, which a service-only correction does not model. Both models are upper bounds, so the reported AoI remains safe rather than optimistic.

Changes. A new queue routine implements the $M/D/c//N$ approximation; the $M/D/c//N$ series was added to the DES-validation figure (Fig. 3) and is discussed in Section VI-B. The conclusion was updated to reflect that this refinement is now included.

Comment 4: Related work, dynamic fleet charging

Reviewer: The paper discusses fleet charging but lacks citations on dynamic fleet charging.

Response. We added a dedicated discussion of dynamic fleet charging with three references: auction-based charging scheduling for a multi-drone network, decentralized electric-vehicle charging coordination, and mobile wireless-charger routing for a sensor fleet. We state precisely what they lack relative to our work: they schedule when and where each vehicle charges, but do not jointly place and size the shared stations, nor couple the induced charging wait into an Age-of-Information objective for a UAV swarm.

Changes. Section II (Related Work), “Energy-limited UAVs and charging”; three references added (now 36 references total).

Comment 5: Writing, machine-generated phrasing

Reviewer: Several phrases read as machine-generated (for example “honest reporting”, “sweet spot”, “not a cosmetic refinement”, and frequent use of “honestly”). The sentence “We stress an honest point: an earlier version ...” suggests a revision of a prior draft.

Response. We revised the prose throughout to remove these phrases. In particular, “sweet spot” is now “optimal swarm size”, “not a cosmetic refinement” is now “essential rather than incidental”, and the words “honest” and “honestly” were removed. The “an earlier version” sentence was rewritten as a hypothetical model comparison (“Had the charging system been modeled as an open $M/M/c$ queue, a spurious crossover would appear ...”), which states the modeling point without referring to any prior draft.

Changes. Sections I, II, and VI.

Comment 6: Equation (12d) exceeds the column width

Response. The constraint (12d) was shortened to a compact form, “ $\text{AoI}_k(t)$ evolves as in (1)”, which fits within the column.

Changes. Section IV, problem (P0).

Comment 7: References are not clean (“Available” links)

Response. We removed the “[Online]. Available: ...” URLs from all published references, which now show only the DOI. The single preprint reference shows a clean “arXiv:XXXX” identifier without a link.

Changes. The reference list and the bibliography source.

Additional strengthening beyond the specific comments

- An *optimality-gap* study benchmarks the greedy block-coordinate solver against the exact optimum (enumerating all placements, port splits, and assignments) on small instances: the solver matches the optimum for $M \leq 6$ and stays within 0.11% on average (below 1% worst case) at $M = 8$. This neutralizes the concern that the greedy solver has no optimality guarantee (Section VI, Table IV).
- A *robustness* study shows the advantage is stable across the collection radius and the charge time, and grows on a non-uniform (clustered) sensor field (Section VI, Fig. 9).

All results are reproducible; the code and data are available at <https://github.com/haodpsut/swarm-uav-aoi-congestion>.

We believe the revision addresses every point raised, and we thank the reviewer again for comments that materially improved the manuscript.

Sincerely,
The Authors